

DISTRIBUTED SYSTEMS AND CONSENSUS ALGORITHMS

Raft Consensus:

Raft is a consensus algorithm designed to be more understandable than Paxos while providing the same guarantees.

Leader Election:

In Raft, a leader is elected when follower nodes experience an election timeout without receiving heartbeats from a leader.

Log Replication:

The leader receives client commands, appends them to its local log, and replicates them to follower nodes.

VECTOR EMBEDDINGS AND HIGH-DIMENSIONAL SEARCH

Vector Embeddings:

Vector embeddings are dense numerical representations of unstructured data such as text, code, or images.

Cosine Similarity:

Cosine similarity measures the cosine of the angle between two non-zero vectors in an inner product space.

Hierarchical Navigable Small World (HNSW):

HNSW is a graph-based indexing algorithm that enables efficient approximate nearest neighbor search in high-dimensional spaces.

DATABASE STORAGE ENGINES AND ACID GUARANTEES

B-Tree Indexing:

B-Trees and B+ Trees maintain sorted key-value entries in balanced hierarchical node structures, supporting efficient range queries and point lookups.

Write-Ahead Logging:

Write-Ahead Logging (WAL) ensures database atomicity and durability by writing transaction mutations to a log before applying them to the data pages.

Two-Phase Commit:

Two-Phase Commit is an atomic commitment protocol that coordinates distributed database transactions across multiple nodes to ensure consistency.